

Data Platform - Pipelines - Interview task (Internship)

Internship Software Engineer Task — Event Log Processor

Overview

Build a small application that processes event logs and produces aggregated statistics.

The goal of this task is not only to produce working code, but also to demonstrate:

- clean code structure
- reasoning and problem solving
- error handling
- testing approach
- ability to make sensible assumptions

You may use AI tools if you wish. However, you should fully understand your implementation and be able to explain your decisions during the follow-up interview.

Functional Requirements

The application must:

1. Read an input file containing one JSON object per line
 2. Parse valid events
 3. Handle malformed input gracefully
 4. Validate event structure and required fields
 5. Produce aggregated statistics based on valid events only
 6. Print the results to the console (or expose them through a simple API if preferred)
-

Event Format

Each line in the file represents a single event.

Example valid events:

```
1 {"timestamp":"2026-05-01T10:00:00Z","eventId":"550e8400-e29b-41d4-  
2 a716-446655440000","userId":"c1b7d8f0-1c3a-4d95-8d0d-  
3 6df3f1d5b001","action":"login"}  
4 {"timestamp":"2026-05-01T10:01:12Z","eventId":"550e8400-e29b-41d4-  
5 a716-446655440001","userId":"c1b7d8f0-1c3a-4d95-8d0d-
```

```
6df3f1d5b001", "action": "view", "articleId": "art-900"}
4
5 {"timestamp": "2026-05-01T10:01:45Z", "eventId": "550e8400-e29b-41d4-
a716-446655440002", "userId": "c1b7d8f0-1c3a-4d95-8d0d-
6df3f1d5b001", "action": "click", "target": "subscribe-button"}
6
7 {"timestamp": "2026-05-01T10:02:05Z", "eventId": "550e8400-e29b-41d4-
a716-446655440003", "userId": "d2d44db8-b8d9-4b43-9c2f-
3bb47e87f221", "action": "purchase", "amount": 19.99}
8
9 {"timestamp": "2026-05-01T10:03:40Z", "eventId": "550e8400-e29b-41d4-
a716-446655440004", "userId": "f2f09c5a-88bc-4c3e-8b78-
0a91cfa0f332", "action": "logout"}
10
```

Supported Actions

The application should support the following event actions:

Action	Description
login	User logged in
logout	User logged out
view	User viewed an article
click	User clicked on a UI element or link
purchase	User made a purchase

Field Requirements

Required Fields For All Events

Field	Type	Description
timestamp	ISO-8601 string	Event timestamp
eventId	UUID	Unique event identifier
userId	UUID	Unique user identifier
action	string	Event action

Additional Fields By Action

view

Field	Type
articleId	string

click

Field	Type
target	string

purchase

Field	Type
amount	decimal number

Input File Example

The input file may contain both valid and invalid lines.

```
1 {"timestamp":"2026-05-01T10:00:00Z","eventId":"550e8400-e29b-41d4-a716-446655440000","userId":"c1b7d8f0-1c3a-4d95-8d0d-6df3f1d5b001","action":"login"}
2
3 {"timestamp":"2026-05-01T10:01:12Z","eventId":"550e8400-e29b-41d4-a716-446655440001","userId":"c1b7d8f0-1c3a-4d95-8d0d-6df3f1d5b001","action":"view","articleId":"art-900"}
4
5 {"timestamp":"2026-05-01T10:01:45Z","eventId":"550e8400-e29b-41d4-a716-446655440002","userId":"c1b7d8f0-1c3a-4d95-8d0d-6df3f1d5b001","action":"click","target":"subscribe-button"}
6
7 {"timestamp":"2026-05-01T10:02:05Z","eventId":"550e8400-e29b-41d4-a716-446655440003","userId":"d2d44db8-b8d9-4b43-9c2f-3bb47e87f221","action":"purchase","amount":19.99}
8
9 INVALID_LINE
10
11 {"timestamp":"2026-05-01T10:03:40Z","eventId":"NOT-A-UUID","userId":"f2f09c5a-88bc-4c3e-8b78-0a91cfa0f332","action":"view","articleId":"art-901"}
12
13 {"timestamp":"2026-05-01T10:04:15Z","eventId":"550e8400-e29b-41d4-a716-446655440005","userId":"INVALID-USER-ID","action":"purchase","amount":5.50}
```

```
14
15 {"timestamp":"2026-05-01T10:05:20Z","eventId":"550e8400-e29b-41d4-
a716-446655440006","userId":"d2d44db8-b8d9-4b43-9c2f-
3bb47e87f221","action":"logout"}
16
17 {"timestamp":"2026-05-01T10:06:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440007","userId":"1c8d5a8f-44f2-49f2-a8aa-
95f4f04d2201","action":"purchase","amount":"invalid"}
18
19 {"timestamp":"2026-05-01T10:06:15Z","eventId":"550e8400-e29b-41d4-
a716-446655440008","userId":"1c8d5a8f-44f2-49f2-a8aa-
95f4f04d2201","action":"unknown"}
20
21 {"timestamp":"2026-05-01T10:07:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440009","userId":"c1b7d8f0-1c3a-4d95-8d0d-
6df3f1d5b001","action":"view","articleId":"art-905"}
22
23 {"timestamp":"2026-05-01T10:08:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440010","userId":"c1b7d8f0-1c3a-4d95-8d0d-
6df3f1d5b001","action":"purchase","amount":13.20}
24
25 {"timestamp":"2026-05-01T10:09:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440011","userId":"d2d44db8-b8d9-4b43-9c2f-
3bb47e87f221","action":"click","target":"article-link"}
26
27 {"timestamp":"2026-05-01T10:10:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440012","userId":"f2f09c5a-88bc-4c3e-8b78-
0a91cfa0f332","action":"login"}
28
29 {"timestamp":"2026-05-01T10:11:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440013","userId":"f2f09c5a-88bc-4c3e-8b78-
0a91cfa0f332","action":"logout"}
30
31 {"timestamp":"2026-05-01T10:12:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440014","userId":"c1b7d8f0-1c3a-4d95-8d0d-
6df3f1d5b001","action":"purchase","amount":7.35}
32
33 {"timestamp":"2026-05-01T10:13:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440015","userId":"8e73f9d7-c2b8-4705-a16f-
dc5dfac98201","action":"view","articleId":"art-903"}
34
35 {"timestamp":"2026-05-01T10:14:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440016","userId":"8e73f9d7-c2b8-4705-a16f-
dc5dfac98201","action":"click","target":"banner-ad"}
36
37 {"timestamp":"2026-05-01T10:15:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440017","userId":"8e73f9d7-c2b8-4705-a16f-
dc5dfac98201","action":"purchase","amount":100.00}
38
39 {"timestamp":"2026-05-01T10:16:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440018","userId":"c1b7d8f0-1c3a-4d95-8d0d-
6df3f1d5b001","action":"logout"}
40
41 {"timestamp":"2026-05-01T10:17:00Z","eventId":"550e8400-e29b-41d4-
a716-446655440019","userId":"d2d44db8-b8d9-4b43-9c2f-
3bb47e87f221","action":"purchase","amount":42.00}
42
```

Validation Rules

A line should be considered invalid if:

- it is not valid JSON

- `eventId` is not a valid UUID
- `userId` is not a valid UUID
- required fields are missing
- the action is unknown
- purchase events contain an invalid amount
- click events are missing `target`
- view events are missing `articleId`
- timestamp is missing or invalid
- `userId` is empty

Invalid lines should NOT stop the program.

The application should continue processing the remaining lines.

Required Statistics

The application must produce the following statistics using valid events only.

1. Total Valid Events

Example:

```
1 | Total valid events: 16
2 |
```

2. Total Invalid Lines

Example:

```
1 | Total invalid lines: 4
2 |
```

3. Event Count Per User

Example:

```
1 | c1b7d8f0-1c3a-4d95-8d0d-6df3f1d5b001: 6
2 | d2d44db8-b8d9-4b43-9c2f-3bb47e87f221: 4
3 | 8e73f9d7-c2b8-4705-a16f-dc5dfac98201: 3
4 |
```

4. Purchase Statistics

Example:

```
1 Total purchase amount: 182.04
2 Average purchase amount: 30.34
3 Largest purchase: 100.00
4
```

5. Most Active User

Example:

```
1 Most active user: c1b7d8f0-1c3a-4d95-8d0d-6df3f1d5b001
2
```

6. Top 3 Most Active Users

Example:

```
1 1. c1b7d8f0-1c3a-4d95-8d0d-6df3f1d5b001 - 6 events
2 2. d2d44db8-b8d9-4b43-9c2f-3bb47e87f221 - 4 events
3 3. 8e73f9d7-c2b8-4705-a16f-dc5dfac98201 - 3 events
4
```

7. Event Count Per Action

Example:

```
1 login: 2
2 logout: 3
3 view: 4
4 click: 3
5 purchase: 5
6
```

Technical Requirements

Language

You may use any OOP programming language.

Application Type

You may implement this as (any of these):

- a console application

- a REST API
 - a small web service
-

Expected Deliverables

Please provide:

1. Source code
 2. Instructions for running the application
 3. A short README explaining:
 - design decisions
 - assumptions made
 - tradeoffs considered
 4. Unit tests
-

Time Expectation

The task is intentionally designed to be small.

Expected effort: approximately: 8 hours of work.

You do not need to complete it in one sitting. We typically expect candidates to finish the task within 1–2 days after receiving it. The deadline is 7 days from receiving it.

Bonus (Optional)

These are optional and not required:

- streaming large files efficiently
 - Docker support
 - concurrency/parallel processing
 - exposing metrics through an API
 - deduplication by `eventId`
 - configurable validation rules
 - CSV or JSON summary output
-

Evaluation Criteria

We are not looking for a perfect solution.

We will evaluate:

- code readability
- structure and maintainability
- correctness
- error handling
- testing approach
- reasoning and tradeoff decisions
- ability to explain the implementation